

PlayIT: Offline-First Gamified Early Literacy Mobile Application

MVP Validation Framework & SMART Evaluation Model (Weeks 1–2 Deliverable)

Course: IT411 — Capstone & Research 2 | Semester 1, AY 2026–2027
Institution: College of Computer Studies, Cebu Institute of Technology – University
Theoretical Frameworks: ISO 9241-11 Usability Model • UPA Framework • Technology Acceptance Model (TAM)
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1. Research Rationale & Validation Objective

Early literacy in Philippine public schools faces large class sizes and limited home connectivity. The Department of Education (DepEd) emphasizes the **Marungko Approach**—a phono-syllabic technique teaching letters by frequency of use. **PlayIT** provides home-based phonetic practice via a 100% offline Android application with Vosk speech recognition and diagnostic telemetry.

The Weeks 1–2 MVP Validation is **formative and diagnostic**: (1) evaluate software usability and child task completion; (2) obtain expert validation on the 26-letter English Marungko progression and 33 CVC Blend It words; and (3) refine evaluation instruments before final summative testing.

2. Tripartite Multi-Stakeholder Evaluation Framework

PlayIT synthesizes three complementary theoretical models across four user cohorts:

- **ISO 9241-11 (Usability Framework):** Evaluates Effectiveness (task success, speech accuracy), Efficiency (time-on-task, retries), and Satisfaction (visual smiley scale).
- **UPA Framework (Usability, Pedagogy, Accessibility):** Evaluates pedagogical fidelity to Marungko, CVC decodability, pediatric accessibility (≥64dp touch targets, 24sp fonts), and non-punitive feedback.
- **Technology Acceptance Model (TAM):** Measures Perceived Usefulness (PU) and Perceived Ease of Use (PEOU) for parents and educators.

3. Curriculum Scope: 26-Letter Marungko English Adaptation

PlayIT adapts the 28-letter Filipino Marungko sequence to **26 letters (A–Z)** for Grade 1 English phonics. The Spanish-derived grapheme **Ñ** does not exist in English. The digraph **NG** (/ŋ/) is a phonogram rather than a single letter, and teaching it as an isolated grapheme causes phonological confusion. The 26 letters are organized into **7 thematic chapters**:

Chapter	Target Letters	Count	Biome Theme	Canonical Blend It Words (CVC Validated)
Chapter 1	m, s, a, i	4	Guava Greenery	SAM, SIS, AIM (3 words — Group 1 exception)
Chapter 2	o, b, e, u	4	Mango Orchard	BUS, SUB, MOM, BEE, BIB
Chapter 3	t, k, l, y	4	Chocolate Hills	BAT, MAT, KIT, TOY, BOY
Chapter 4	n, g, p	3	Palm Valley	PIG, PAN, BUG, PIN, NAP
Chapter 5	r, d, h, w	4	Rice Terraces	DOG, HAT, HEN, BED, WEB
Chapter 6	c, f, j	3	Coral Reef	CAT, FAN, CAP, CUP, JAM
Chapter 7	q, v, x, z	4	Mt. Pulag Summit	VAN, BOX, FOX, ZOO, QUIZ

All 33 words are constraint-validated — each word uses only letters available from current and prior chapters. This table is the canonical design target; the APK database will be updated to match these word banks prior to field testing.

4. SMART Evaluation Objectives & Measurable Metrics

SMART Dimension	Project Objective	Evaluation Metric	Target Criteria	Cohort
Specific (S)	Verify learner task success across Hear It, Say It, Find It, Blend It.	Sublevel completion rate; 1st-attempt vocal accuracy; time-on-task.	>=85% completion rate; >=75% ASR accuracy.	Early Learners (N=12)
Measurable (M)	Quantify caregiver usability and technology acceptance.	SUS score; 5-point Likert TAM scales (PU, PEOU, BI).	Mean SUS >=75.0; TAM Perceived Usefulness >=4.2/5.0; Behavioral Intention >=4.0/5.0.	Parents & Guardians (N=10)
Achievable (A)	Validate pedagogical sequence with certified educators.	Expert Review Rubric (7 items, 1-5 scale).	100% agreement (>=4.0/5.0) on Marungko sequence and decodable words.	DepEd Teachers & SMEs (N=5)
Relevant (R)	Ensure 100% offline persistence, data privacy, and diagnostic reporting.	Room DB persistence audit; PDF export time; Noise gate test.	0% data loss on force-kill; PDF export <=3.0s; Noise alert at >40dB.	Technical Evaluators (N=3)
Time-Bound (T)	Execute validation, analyze findings, and refactor SRS/SDD.	Submission milestone dates per IT411 calendar.	Validation Protocol: Sept 12; User Trials: Sept 18; Refactored SRS/SDD: Sept 19.	Research Team (Total N=30)

5. Stakeholder Mapping & Questionnaire Item Cross-Walk

Code	Evaluation Criterion / Question Item	Construct	Target Respondent	Scale
PED-01	Strictly follows DepEd Marungko frequency-based progression.	UPA: Pedagogy	Teachers / SMEs	1–5 Likert
PED-02	Hear It delivers clear, natural phoneme sound modeling.	UPA: Audio Quality	Teachers / SMEs	1–5 Likert
PED-03	Say It uses age-appropriate example words (Mouse, Sun).	UPA: Vocal Practice	Teachers / SMEs	1–5 Likert
PED-04	Find It uses recognizable, culturally familiar images.	UPA: Visual Design	Teachers / SMEs	1–5 Likert
PED-05	Blend It uses decodable CVC words (e.g., MOM, KIT, PIG, HEN, CAP).	UPA: Word Blending	Teachers / SMEs	1–5 Likert
PED-06	Excluding NG and N-tilde to align with 26-letter English phonics is sound.	UPA: Scope Validity	Teachers / SMEs	1–5 Likert
PED-07	Non-punitive gamification encourages learner persistence.	UPA: Motivation	Teachers / SMEs	1–5 Likert
TAM-PU01	Helps child learn letter sounds independently at home.	TAM: Usefulness	Parents / Guardians	1–5 Likert
TAM-PU02	App makes practice more engaging than paper worksheets.	TAM: Usefulness	Parents / Guardians	1–5 Likert
TAM-PEO U01	Easy for child to navigate without constant adult supervision.	TAM: Ease of Use	Parents / Guardians	1–5 Likert
TAM-PEO U02	Creating a profile and selecting an avatar was quick and intuitive.	TAM: Ease of Use	Parents / Guardians	1–5 Likert
TAM-BI01	I intend to use PlayIT regularly with my child for phonics practice at home.	TAM: Behavioral Intention	Parents / Guardians	1–5 Likert
DASH-01	Parent Dashboard clearly shows mastered sounds vs practice needs.	TAM: Utility	Parents / Guardians	1–5 Likert
DASH-02	Math question gate prevents accidental child entry to settings.	ISO: Security	Parents / Guardians	1–5 Likert
OFF-01	100% offline capability (no WiFi/data) is practical and cost-saving.	TAM: Practicality	Parents / Guardians	1–5 Likert
SAFE-01	Safe learning environment with zero ads or paid purchases.	ISO: Child Safety	Parents / Guardians	1–5 Likert
OBS-01	Child independently selects profile and taps active node on map. (Time recorded)	ISO: Effectiveness + Efficiency	Child (Observed)	3-pt + Time(s)
OBS-02	Child attends to Hear It audio model and repeats target sound.	UPA: Engagement	Child (Observed)	3-pt Rubric
OBS-03	Child presses mic button and produces verbal phonics response. (Time recorded)	ISO: Usability + ASR Efficiency	Child (Observed)	3-pt + Time(s)
OBS-04	Child correctly identifies 3 target picture cards in Find It.	ISO: Discrimination	Child (Observed)	3-pt Rubric
OBS-05	Child arranges letter tiles into correct CVC word in Blend It. (Time recorded)	UPA: Synthesis + Efficiency	Child (Observed)	3-pt + Time(s)
SMILEY	Child post-session affective reaction (Sad = 1, Neutral = 2, Happy = 3).	ISO: Satisfaction	Early Learner	3-pt Visual
SYS-01	Speech recognition latency from mic release to feedback is <=0.5s.	ISO 25010: Latency	Technical Evaluator	Pass / ms
SYS-02	Application maintains 60 FPS animation stability without stutter.	ISO 25010: Smoothness	Technical Evaluator	Binary Pass
SYS-03	Room DB persists progress and stars across force-close restarts.	ISO 25010: Integrity	Technical Evaluator	Binary Pass

6. Validation Administration Protocol & Academic Evidence

Field evaluations will be conducted during Week 2 using the pre-packaged playit-debug.apk running on 5 physical Android devices. Observations and survey submissions will be synchronized to the master Google Sheet. The empirical findings will be compiled into the **MVP Validation Highlights Report** and directly substantiate the **SRS v3.0 Requirements Traceability Matrix (RTM)** and **SDD v2.0** refactoring due in Week 3.